3. *(a)* Crude oil can be separated into simpler mixtures called fractions. These fractions contain hydrocarbon compounds called alkanes. **Table 1** shows information about some of the fractions obtained from crude oil by fractional distillation.

Fraction	Boiling point range (°C)	Number of carbon atoms present in the alkanes
petroleum gases	< 20	C_1-C_4
petrol	30-75	C_5-C_{10}
naphtha	70-170	C_8-C_{12}
kerosene	170-250	C_{10}-C_{14}
diesel oil	250-340	C_{14}-C_{24}
lubricating oil	340-500	C_{21}-C_{30}
fuel oil	490-580	C_{25}-C_{35}
residue	> 580	$>C_{35}$

Table 1

Use **only** the information in **Table 1** to answer parts (i)-(iii).

(i) Hexane has a boiling point of 68 °C. Give the name of the fraction which contains hexane. [1]

....

(ii) One alkane is found in kerosene and in diesel. Give the number of carbon atoms in this alkane. [1]

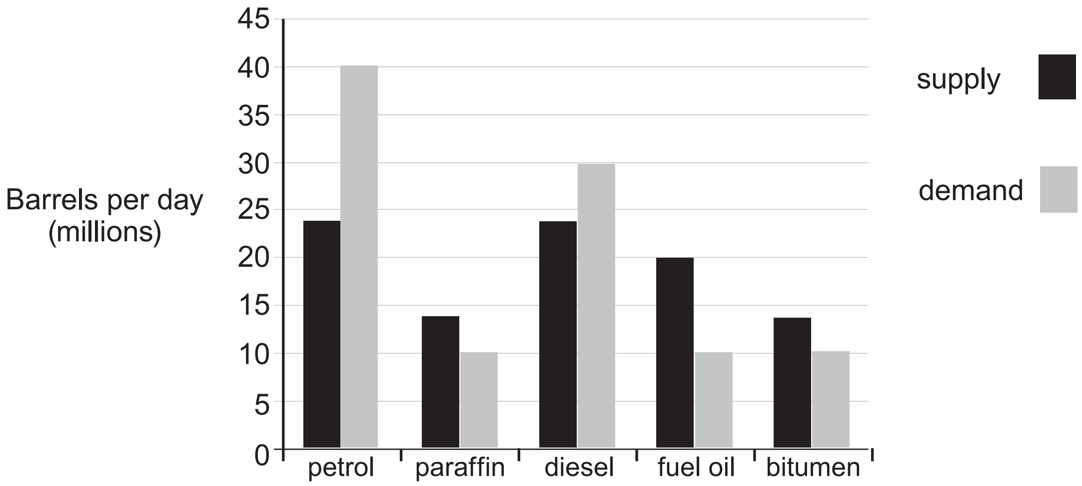
....

(iii) Give the number of carbon atoms in the alkane which has the **lowest** boiling point. [1]

....

Examiner only

(b) The bar chart shows the **supply** and **demand** for some crude oil products.



Name the fractions for which the demand is greater than the supply. Suggest a reason why these fractions are in high demand. [2]

Fractions .... and ....

Reason ....

....

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